Visual Line Tracking (No-PTZ)

Experiment Objective

The camera recognizes ground lines (HSV color segmentation), PID controls the chassis angular velocity to drive along the line, and the LiDAR stops the car on obstacles ahead. The No-PTZ version uses a fixed camera bracket without the pan-tilt tilt-down function.

Experiment Principle

1. Keyword Explanation

Keyword	Description
Line Following	The camera recognizes the line position, uses the image center as the reference to compute the error, and PID closed-loop controls the angular velocity
HSV Color Segmentation	Uses `cv.inRange` in the HSV color space to binarize and extract lines of a specific color, with hue wrap-around support (e.g., red H crossing 0°)
Line Tracking ROI	Only processes the lower part of the image (`line_roi_top_ratio`=0.55), excluding distant background interference
Discrete PID Controller	Independent P+I+D gains with integral clamping; pixel error is scaled by `line_error_divisor` and normalized by `pid_scale`(1000)
Laser Obstacle Avoidance Stop	If valid points in the 0.20~0.35m range within the front ±90° half-angle of `/scan` exceed 10 → immediate stop + optional buzzer
Automatic Exit on Line Loss	Automatically calls `rclpy.shutdown()` after the line-lost timeout `line_lost_timeout_sec`(1.5s)
Mouse HSV Sampling	Selects a ROI with the mouse → takes the 5~95 percentile HSV → automatically updates the threshold → saves to a file
QoS	best_effort (image/laser input + cmd_vel downlink), sensor_data (LiDAR)
TF Coordinate Frames	odom → base_footprint → base_link

2. Technical Architecture

Dual timers (image 30Hz + control 20Hz) + laser callback + discrete PID controller:

```
/camera/image_raw (BEST_EFFORT)
  → follow_line_node
    ├── Image processing timer (30Hz): image conversion → flip → HSV binarization
→ morphology → largest contour → EMA filtering
    └── Chassis control timer (20Hz): pixel error → scaling → discrete PID →
angular velocity clamping → /cmd_vel (BEST_EFFORT)
/scan (sensor_data)
  → Obstacle avoidance callback: count valid front points → above threshold →
stop + buzzer
/JoyState (Bool)
  → Joystick override → stop + PID reset
```

Differences between No-PTZ and PTZ versions: the camera is fixed and cannot tilt down (no pan-tilt). Compensate by manually adjusting the camera mounting angle, or increasing `line_roi_top_ratio` to enlarge the lower ROI.

3. Core Topics Quick Reference

Firmware Uplink (Sensor Data)

Topic	Message Type	QoS	Description
`/camera/image_raw`	`sensor_msgs/Image`	best_effort	ESP32 camera raw image frames
`/scan`	`sensor_msgs/LaserScan`	sensor_data	LiDAR scan data (for obstacle avoidance)

Firmware Downlink (Control Commands)

Topic	Message Type	QoS	Field	Range	Direction
`/cmd_vel`	`geometry_msgs/Twist`	best_effort + keep_last(1)	`linear.x`	0.15 m/s (constant)	`+=` forward
			`angular.z`	-1.20~1.20 rad/s	`+=` turn left, `-=` turn right
`/beep`	`std_msgs/UInt16`	reliable	`data`	0/1	1=buzzer on (obstacle avoidance triggered)

Usage example:

```
# Increase line tracking speed
ros2 launch microros_v2_no_ptz_visual_control_pkg follow_line.launch.py \
    line_linear_speed:=0.20 max_angular_speed:=1.5
```

ROS2 Internal Topics

Topic	Message Type	QoS	Source	Description
`/JoyState`	`std_msgs/Bool`	reliable	Joystick node	When the joystick is active, follow_line_node stops + PID reset

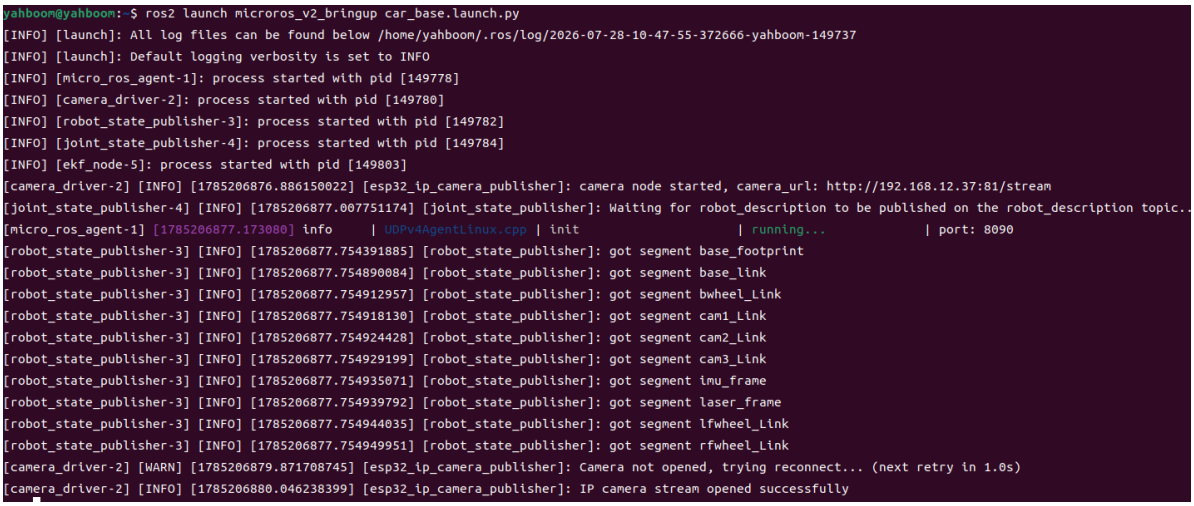
Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	❌ This section is for the No-PTZ version
No-PTZ version	☑
Required peripherals	ESP32 camera (fixed bracket) + T-mini Plus LiDAR

2. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```



3. Start Visual Line Tracking (Terminal 2)

```
ros2 launch microros_v2_no_ptz_visual_control_pkg follow_line.launch.py
```

5. Operation Instructions

Interaction

Key	Function
Space	tracking/paused
i / I	Recognition only (does not control the chassis)
r / R	Reset
Mouse drag	HSV sampling (drag to select the line area, threshold auto-updates)
q / Q / ESC	Quit

Safety and Boundary Handling

- Obstacle avoidance takes priority over the line tracking PID
- Joystick override → stop + PID reset
- Deadband: drive straight within 40px (PID reset)
- Line lost 1.5s → stop + optional automatic exit
- PID integral clamping prevents saturation (`pid_integral_limit`=100)
- Hue wrap-around handles red lines (two-segment mask merged when Hmin > Hmax)
- Stop before quitting

Experiment Result

- **Automatic line tracking**: drives along the line at 0.15 m/s, PID corrects deviations
- **Line centered**: drives straight within the 40px deadband

- **Line to the left**: turns right to correct; turns left when the line is to the right
- **Obstacle ahead**: stops + buzzer within 0.35m, resumes after the obstacle is removed
- **Line lost >1.5s**: automatically stops and exits
- **Joystick**: stops, resumes after release

Code Explanation

> Source paths:
>
> - Launch: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/follow_line.launch.py`
> - Node source: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/follow_line_node.py`

1. Core Node Analysis

`follow_line_node` is the most feature-complete of the 4 visual control nodes — integrating HSV line tracking, discrete PID control, laser obstacle avoidance, mouse sampling, and joystick override. It uses a dual-timer architecture to decouple image processing from chassis control.

Data flow:

- `image_callback` → only caches the latest raw `Image` (non-blocking)
- `process_latest_image` (30Hz) → image conversion → orientation correction → ROI HSV binarization → largest contour extraction → EMA filtering → state update
- `publish_chassis_control_command` (20Hz) → pixel error `/ line_error_divisor` → discrete PID → angular velocity clamping → publish `/cmd_vel`
- `scan_callback` → count valid front points → obstacle avoidance stop when above the threshold

Constructor — declares parameters, creates the discrete PID controller, subscribers/publishers/dual timers:

```
# Source file:
~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/follow_line_node.py

def __init__(self) -> None:
    """Initialize the line following node."""
    super().__init__(DEFAULT_NODE_NAME)

    self.image_lock = threading.Lock()
    self.line_state_lock = threading.Lock()
    self.scan_state_lock = threading.Lock()

    # Cache the latest image (does not block the subscription callback)
    self.latest_image_msg: Optional[Image] = None
    self.latest_image_sequence_id = 0
    self.processed_image_sequence_id = 0

    self.tracking_state = TRACKING_STATE_IDENTIFY
    self.is_node_running = True
    self.joy_is_active = False
```

```
    # Line detection state
    self.line_is_valid = False
    self.raw_line_center: Optional[Tuple[float, float]] = None
    self.filtered_line_center: Optional[Tuple[float, float]] = None  # After EMA
filtering
    self.latest_line_update_time_sec = 0.0
    self.line_lost_start_time_sec: Optional[float] = None

    # Obstacle state
    self.obstacle_warning_count = 0
    self.obstacle_is_active = False

    # Mouse HSV sampling state
    self.mouse_selecting = False
    self.roi_selection_start: Tuple[int, int] = (0, 0)
    self.roi_selection_end: Tuple[int, int] = (0, 0)
    self.roi_selection_pending = False

    self.declare_and_get_parameters()
    self.load_hsv_parameters_from_file_if_needed()

    # Create the discrete PID controller (gains normalized by pid_scale)
    self.line_pid_controller = DiscretePidController(
        self.line_pid_kp / self.pid_scale,   # 22.0 / 1000 = 0.022
        self.line_pid_ki / self.pid_scale,   # 0.0
        self.line_pid_kd / self.pid_scale,   # 5.0 / 1000 = 0.005
        self.pid_integral_limit,             # 100.0
    )

    # Image subscription (QoS: KEEP_LAST, BEST_EFFORT)
    self.image_subscription = self.create_subscription(
        Image, self.image_topic, self.image_callback,
        QoSProfile(history=QoSHistoryPolicy.KEEP_LAST, depth=1,
                   reliability=QoSReliabilityPolicy.BEST_EFFORT,
                   durability=QoSDurabilityPolicy.VOLATILE),
    )
    # LiDAR subscription (sensor_data QoS)
    self.scan_subscription = self.create_subscription(
        LaserScan, self.scan_topic, self.scan_callback, qos_profile_sensor_data,
    )
    # Joystick subscription
    self.joy_state_subscription = self.create_subscription(
        Bool, self.joy_state_topic, self.joy_state_callback, 10,
    )

    # cmd_vel publisher (BEST_EFFORT aligned with firmware)
    self.cmd_vel_qos = QoSProfile(
        history=QoSHistoryPolicy.KEEP_LAST, depth=1,
        reliability=QoSReliabilityPolicy.BEST_EFFORT,
        durability=QoSDurabilityPolicy.VOLATILE,
    )
    self.cmd_vel_publisher = self.create_publisher(Twist, self.cmd_vel_topic,
self.cmd_vel_qos)
    self.beep_publisher = self.create_publisher(UInt16, self.beep_topic, 10)

    # Dual timers
    self.image_processing_timer = self.create_timer(
        1.0 / max(self.image_processing_rate_hz, MINIMUM_TIMER_RATE_HZ),
```

```
        self.process_latest_image,
    )
    self.control_timer = self.create_timer(
        1.0 / max(self.control_rate_hz, MINIMUM_TIMER_RATE_HZ),
        self.publish_chassis_control_command,
    )

    self.dynamic_parameter_callback_handle =
self.add_on_set_parameters_callback(
        self.on_dynamic_parameter_update
    )

    if self.auto_start_tracking:
        self.tracking_state = TRACKING_STATE_TRACKING
```

Chassis control timer callback — PID line tracking + obstacle avoidance priority:

```
# Source file:
~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visua
l_control_pkg/follow_line_node.py

def publish_chassis_control_command(self) -> None:
    """Publish chassis control command at a fixed rate."""
    if not self.is_node_running:
        return

    # Joystick override: zero velocity + PID reset (already handled in
joy_state_callback)
    if self.joy_is_active:
        self.publish_zero_twist()
        return

    # Obstacle avoidance priority: obstacle ahead → stop + buzzer
    with self.scan_state_lock:
        local_obstacle_is_active = self.obstacle_is_active
    if self.enable_obstacle_stop and local_obstacle_is_active:
        self.publish_zero_twist()
        self.publish_beep(True)
        return
    self.publish_beep(False)

    if self.tracking_state != TRACKING_STATE_TRACKING:
        if self.tracking_state in (TRACKING_STATE_LOST, TRACKING_STATE_PAUSED)
or self.lost_line_stop:
            self.publish_zero_twist()
        return

    with self.line_state_lock:
        local_line_is_valid = self.line_is_valid
        local_filtered_center = self.filtered_line_center
        local_image_width = self.image_width
        local_latest_update_time_sec = self.latest_line_update_time_sec

    # Line-lost timeout check
    if (not local_line_is_valid or local_filtered_center is None
            or local_image_width <= 0
```

```
            or (time.perf_counter() - local_latest_update_time_sec) >
self.line_lost_timeout_sec):
        self.line_pid_controller.reset()
        self.publish_zero_twist()
        return

    # Pixel error → PID angular velocity
    line_center_x = local_filtered_center[0]
    image_center_x = local_image_width * 0.5
    pixel_error_x = line_center_x - image_center_x

    twist_command = Twist()
    twist_command.linear.x = float(self.line_linear_speed)  # Constant linear
velocity

    if abs(pixel_error_x) <= self.line_center_deadband_pixel:
        angular_command = 0.0                    # Drive straight within the
deadband
        self.line_pid_controller.reset()
    else:
        pid_input_value = pixel_error_x / self.line_error_divisor  # Error
scaling
        pid_output_value = self.line_pid_controller.update(pid_input_value, 0.0)
        angular_command = -pid_output_value if self.reverse_angular_direction
else pid_output_value
        angular_command = float(np.clip(angular_command, -
self.max_angular_speed, self.max_angular_speed))

    twist_command.angular.z = angular_command
    self.cmd_vel_publisher.publish(twist_command)
```

Key logic:

- **HSV hue wrap-around**: when `Hmin > Hmax` (e.g., red: Hmin=0, Hmax=9), `create_hsv_mask()` automatically merges two mask segments (0~Hmax + Hmin~179)
- **EMA filtering**: `target_smooth_alpha` (0.55) applies an exponential moving average to the detected center, reducing jitter
- **Mouse sampling**: `on_mouse_event` records the selected ROI → `update_hsv_from_selected_roi` takes the 5~95 percentile HSV → automatically writes to the parameter server
- **Discrete PID**: `DiscretePidController` is an independent class with independent P/I/D terms + integral clamping, supporting dynamic gain updates

2. Key Parameters

> Parameter definition files:
>
> - Launch: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/follow_line.launch.py`
> - Node: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/follow_line_node.py`

Line Tracking Control Parameters

Parameter	Type	Default	Description
`line_linear_speed`	float	`0.15` m/s	Constant linear velocity for line tracking
`max_angular_speed`	float	`1.2` rad/s	Maximum angular velocity clamping
`line_pid_kp`	float	`22.0`	PID P gain (actual 0.022 after scaling by pid_scale=1000)
`line_pid_ki`	float	`0.0`	PID I gain
`line_pid_kd`	float	`5.0`	PID D gain (actual 0.005 after scaling)
`pid_scale`	float	`1000.0`	PID gain normalization divisor
`pid_integral_limit`	float	`100.0`	Integral clamping
`line_error_divisor`	float	`16.0`	Pixel error scaling divisor
`line_center_deadband_pixel`	float	`40.0` px	Deadband (drive straight when the error is within this range, PID reset)
`reverse_angular_direction`	bool	`true`	Reverse the angular velocity direction

Line Tracking Detection Parameters

Parameter	Type	Default	Description
`line_roi_top_ratio`	float	`0.55`	ROI top ratio (only processes the lower part of the frame)
`line_roi_bottom_ratio`	float	`2.00`	ROI bottom ratio
`line_min_area`	float	`150.0` px²	Minimum contour area
`line_lost_timeout_sec`	float	`1.5` s	Line-lost timeout
`target_smooth_alpha`	float	`0.55`	EMA filter coefficient
`Hmin/Smin/Vmin`	int	`0/85/126`	HSV lower bound
`Hmax/Smax/Vmax`	int	`9/253/253`	HSV upper bound
`morph_kernel_size`	int	`5`	Morphology kernel size

Obstacle Avoidance Parameters

Parameter	Type	Default	Description
`enable_obstacle_stop`	bool	`true`	Laser obstacle avoidance switch
`obstacle_min_valid_distance`	float	`0.20` m	Minimum valid distance
`obstacle_response_distance`	float	`0.35` m	Obstacle response distance
`obstacle_check_angle_deg`	float	`90.0` °	Obstacle check half-angle
`obstacle_warning_count_threshold`	int	`10`	Minimum valid point count to trigger obstacle avoidance

3. Node Description

Node	Package	Description
`follow_line_node`	`microros_v2_no_ptz_visual_control_pkg`	Visual line tracking + obstacle avoidance, dual timers + discrete PID + mouse sampling

Common Problems

Problem	Cause	Solution
Line tracking sways left and right	PID gains too large	Reduce `line_pid_kp` or `line_pid_kd`
Line not detected	HSV thresholds do not match	Select the line area with the mouse to re-sample; check the binary image in `show_binary_window`
Obstacle avoidance too sensitive	`obstacle_response_distance` too large or `obstacle_warning_count_threshold` too small	Reduce the response distance or increase the point count threshold
Sudden stop	`lost_line_stop` stops the car when the line is lost	Set it to `false` as needed, or check the HSV/LiDAR inputs